

Sternoclavicular — Protraction(+)/retraction(-)	Sternoclavicular — Depression(+)/elevation(-)	Sternoclavicular — Backwards(+)/forward(-) rotation
not modelled	not modelled	not modelled
Acromioclavicular — Protraction(+)/retraction(-)	Acromioclavicular — Medial(+)/lateral(-) rotation	Acromioclavicular — Posterior(+)/anterior(-) tilt
not modelled	not modelled	not modelled
Scapulothoracic — Protraction(+)/retraction(-)	Scapulothoracic — Medial(+)/lateral(-) rotation	Scapulothoracic — Posterior(+)/anterior(-) tilt
single group · K= 7	single group · K= 5	single group · K= 6
Glenohumeral — Plane of elevation	Glenohumeral — Elevation(-)/Depression(+)	Glenohumeral — Int(+)/ext(-) rotation
single group · K= 8	single group · K= 5	single group · K= 6

stars = LIKELIHOOD-RATIO TEST of H0 'no ex-vivo departure' against H1, Bonferroni-adjusted across every compared cell on the plate: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns = not significant.

ex/in · studies ex/in | K, this cell's own spline df chosen by BIC | mean and max $|\Delta|$ in degrees. Curves span the full abscissa — every x any shoulder reaches is fitted, so the thinnest stretches rest on a single shoulder and the 95%